

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
1.	(a)	(i)		20 000 = 2 marks Award 1 mark for 2000 X 10(dilution factor) 2000		2		2	2	2
		(ii)		60 minutes		1		1		2
	(b)	(i)		Stationary	1			1		
		(ii)		Counts dead cells (as well as live cells)/ it is a total count/ it is not a viable count	1			1		1
		(iii)		Any four (x1) from A. Take sample/ transfer (a volume) of {culture/ bacteria/ <i>E.coli</i> } (1) B. Serial dilution / description of (1) C. Plate out /description of adding {dilution/owtte} to {agar/ growth medium} (1) Ignore put on/ in a Petri dish D. Incubate /description of being left for (suitable) time (at suitable temperature) (1) E. Count (bacterial) <u>colonies</u> (1) Reject count bacteria		4		4		4
	(c)	(i)		Spherical Reject round/ circular	1			1		
		(ii)		Gram negative + has lipopolysaccharide (membrane)		1		1		

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	(d)	(i)		(Image 1.2 shows a) Gram negative result (1) Reject positive/ stain pink Both {gram negative/ stain pink} (1) Both are Gram negative= 2 marks Both stain pink = 1 mark			2	2		
		(ii)		Syphilis/ <i>Treponema (pallidum)</i> (1) cells are {helical/spiral/coiled/ spirillum} / cells are not spherical (1)			2	2		
				Question 1 total	3	8	4	15	2	9